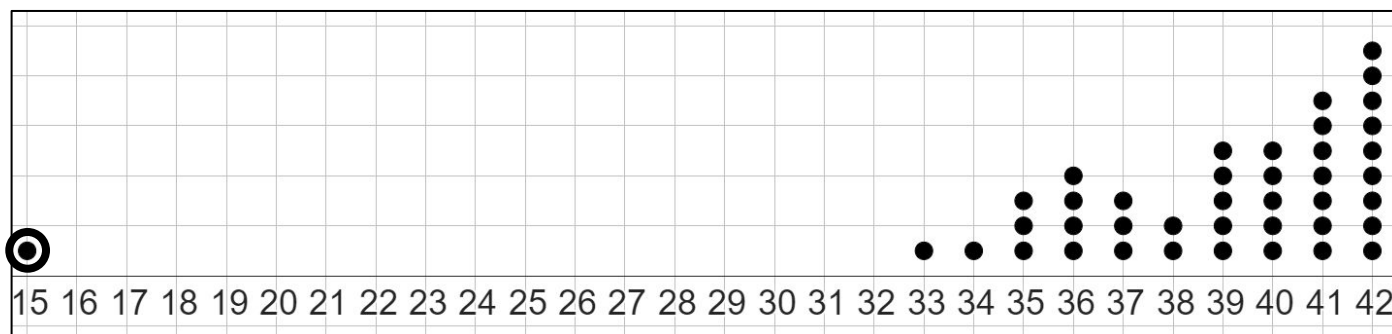




Evaluating the range

Task A: Outliers and their effect on the range

1)



a) Circle the **outlier** in this dataset.

b) What is the original range of this dataset? *The range is 27*

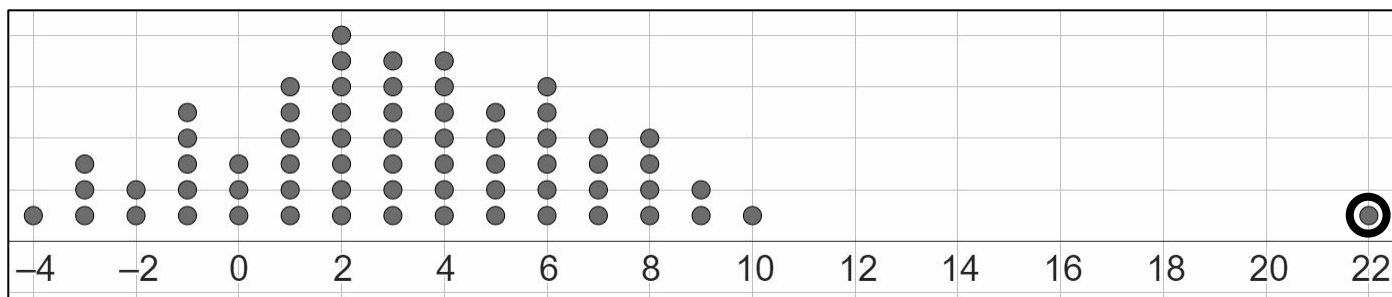
c) If a data point for the number 39 was removed, by how much would the range change? Explain your answer. *Would not change, as 39 is neither the smallest or largest data point.*

d) Remove the **outlier** from the dataset. Calculate the new range.

$$42 - 33 = 9$$

The new range is 9

2)



a) Circle the **outlier** in this dataset.

b) What is the original range of this dataset? *The range is 26*

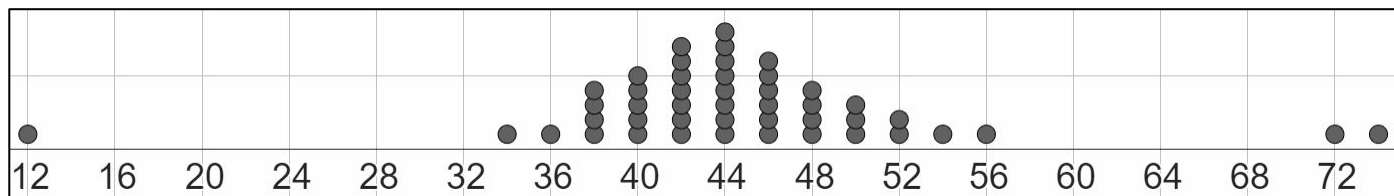
c) If a data point for the number -4 was removed, by how much would the range change? Explain your answer.

The range would decrease by 1 to 25

This is because -4 is the smallest data point.

d) Remove the **outlier** from the dataset. Calculate the new range.

$$10 - (-4) = 14 \quad \text{The new range is 14}$$



3) All data points in this dataset are even numbers.

a) What are the values of the three **outliers**?

12, 72, 74

b) By how much would the range decrease by if all the **outliers** were removed?

The range is currently 62

Without the three **outliers**, the range would become:

$$56 - 34 = 22$$

The range would decrease by 40

1.5	14.5	16.9	21	24.2	28.3	31.9	33.8	37.7	38.8	39.9	42.5	45.2	46.8	82.5
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4) Complete each sentence with a number from the dataset:

- the original range is 81
- removing the data point 37.7 will change the new range to 81
- if the smaller **outlier** was removed from the original dataset, the new range would be 68
- if the larger **outlier** was removed from the original dataset, the new range would be 45.3
- the **outlier** that would be more impactful to remove is the larger outlier: 82.5

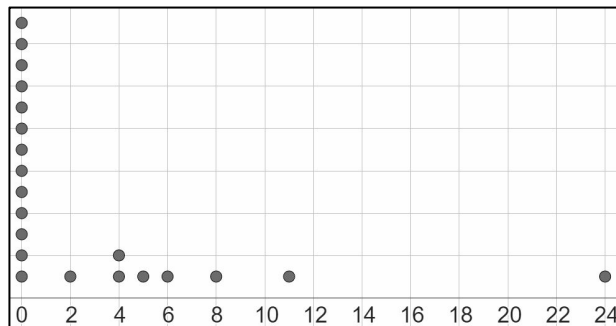


Task B: Evaluating if removing outliers is sensible

1) This dot plot shows the number of days per month that had airfrost (due to cold weather) in Leuchars (Scotland) over a 21 month period.

a) What is the value of the **outlier**?

24 is the **outlier**



b) Should the **outlier** be: removed, kept, or investigated further?

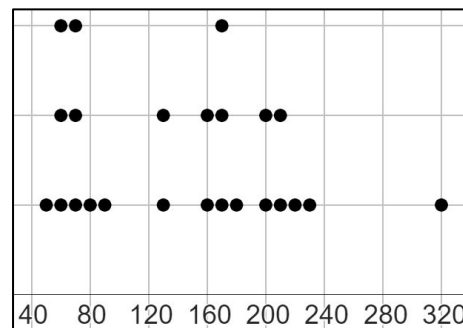
Write a sentence to explain the option you have chosen.

*Investigate further, as it is unclear whether the **outlier** is an error, or just a very cold month.*

2) This dot plot shows the number of “hours of sunshine” per month for two years, rounded to the nearest 10 hours.

a) What is the value of the **outlier**?

320 is the **outlier**



b) All data points have been correctly measured, with no error.

Should the **outlier** be: removed, kept, or investigated further?

Write a sentence to explain the option you have chosen.

*The **outlier** should be kept, as this month might have just been a very sunny month, and this information needs to be represented.*

3) This dataset shows the height of 24 chimpanzees, in centimetres.

a) What is the current range of this dataset?

196.6 cm

1.4	95	103	108	108	111	114	117
118	118	124	125	129	132	135	138
130	138	150	153	161	166	172	198

b) What is the value of the **outlier** or **outliers** in this dataset?

1.4 is an **outlier**

*198 may be an **outlier**, but this is unclear.*

c) Should the **outlier** or **outliers** be: removed, kept, or investigated further?

Write a sentence to explain the option you have chosen, and justify it by calculating any new range that might happen from you removing any **outliers**.

*The **outlier** of 1.4 should be removed, as this is very clearly an error. Perhaps this measurement was made in metres, not centimetres. The range would decrease to 103. The data point of 198 needs to be investigated further, it might just be a very large chimpanzee!*



4) This dataset accurately shows the total monthly rainfall (in millimetres) each month for two years, between 2007 and 2008.

Jun wants to represent this dataset on a dot plot, and wants to see if removing any **outliers** has an impact on the range of the data he has to plot.

Calculate two ranges, with one keeping the **outliers**, and one removing an **outlier**, and suggest if Jun should represent all data points on a dot plot or not.

285.6	60
154	58.3
146.9	57.5
113	56.7
108.7	52.8
98.8	48.7
93.2	44.5
92.4	43.5
83.8	28.1
82.6	27.1
79.2	23.3
74.9	5.8

original range = 279.8

*If the **outlier** of 285.6 is removed from the dataset, the new range becomes 148.2*

*Jun should omit the **outlier** from the dot plot, as the range nearly halves without it. However, the **outlier** should still be acknowledged, in case that month was just very rainy!*

